

Quiz: CPU and Addressing

This quiz has 5 questions. The passing grade is 4.

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Started on	Tuesday, January 14, 2025, 8:39 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 8:54 PM
Time taken	15 mins 5 secs
Grade	4.00 out of 5.00 (80%)

Question **1**

Correct

1.00 points out of 1.00

Long displacement is 20 bits long

Select one:

- ☒ True ✓
- ☐ False

Correct!

The correct answer is 'True'.

Question 2

Incorrect

0.00 points out of 1.00

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Given the base-displacement addressing halfword CFFF. If register 12 contains hexadecimal 00000000 00040000, the effective address is

Select one:

- ☒ A. 00000000 0000000C ✖
- ☐ B. 0000000C 00000FFF
- ☐ C. 00000000 00040FFF
- ☐ D. 00040000 00000FFF
- ☐ E. 00000000 0003FFFF

The base register, C (decimal 12), contains hexadecimal 00000000 00040000.

The displacement is hexadecimal FFF, or decimal 4095, bytes from the base. (Note it is not a negative displacement. Unlike long displacement or relative addressing, 12-bit displacements are unsigned.)

The effective address is $00000000\ 00040000 + 00000000\ 00000FFF = \mathbf{00000000\ 00040FFF}$.

The correct answer is: 00000000 00040FFF

Question **3**

Correct

1.00 points out of 1.00

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If a relative address operand is hexadecimal 000A, it indicates

Select one:

- ☒ A. 20 bytes after the instruction being executed ✓
- ☐ B. 10 bytes preceding the instruction being executed
- ☐ C. 5 bytes after the instruction being executed
- ☐ D. 10 bytes after the instruction being executed
- ☐ E. 20 bytes preceding the instruction being executed

Correct!

Relative address operand units are halfwords. In this case, 000A, 10 halfwords, is 20 bytes. The number is positive, so it refers to 20 bytes after the instruction.

The correct answer is: 20 bytes after the instruction being executed

Question **4**

Correct

1.00 points out of 1.00

The CPU has a set of registers, which are a very fast form of memory

Select one:

- ☒ True ✓
- ☐ False

Correct!

The correct answer is 'True'.

Question **5**

Correct

1.00 points out of 1.00

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Given the base-displacement addressing halfword 0020: if register 0 contains hexadecimal 00000000 00040000, the effective address is

Select one:

- ☒ A. 00000000 00000020 ✓
- ☐ B. 00000000 00040020
- ☐ C. 00000000 00040000
- ☐ D. 00000020 00040000
- ☐ E. 00040020 00040020

Correct!

Register 0 is never used as base. If specified in an addressing halfword, it is ignored.

When the base in the addressing halfword is 0, the effective address is just the displacement.

The correct answer is: 00000000 00000020